IRT-FIT: Detecting Benchmark Contamination in Large Language Models via Person-Fit Statistics at the Minimal-Information Operating Point

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Abstract

Benchmark contamination undermines the reliability of Large Language Model (LLM) evaluation, yet all existing detection methods require privileged access: training data, token probabilities, model weights, or specially designed prompts. We establish the *minimal-information operating point* on the contamination detection frontier by repurposing Item Response Theory (IRT) person-fit statistics—developed over decades for detecting exam cheating—to LLM benchmark integrity. Using only binary (correct/incorrect) response patterns from the target model, together with externally calibrated 4-Parameter Logistic (4PL) item parameters, our framework IRT-FIT detects controlled contamination with |Cohen's d| $>$ 2.9 across three model families (Qwen, Llama, Mistral), achieving $8\times$ the effect size of classical person-fit indices (Guttman/U3). The simpler HT statistic achieves comparable model-level detection power; however, ℓz^* uniquely provides calibrated hypothesis testing via a known $\mathcal{N}(0,1)$ null, Snijders correction for ability estimation error, and full 4PL item-heterogeneity weighting. The person-fit family retains detectable signal after reinforcement learning ($|d| > 1.3$), where all token-probability methods collapse to chance (AUROC ≤ 0.52). We further show that 4PL parameterization is essential (3PL loses $\sim$27pp detection power on average), detection power scales with benchmark size (97.9% for MMLU-scale, 12.9% for ARC-C), and domain-level analysis reveals distinct contamination typologies. Our IRT parameter recovery analysis quantifies the sample-size gap between classical and neural IRT estimation, charting a path toward large-scale retrospective audits.

1 Introduction

As LLMs are increasingly evaluated on standardized benchmarks, contamination—the inclusion of benchmark items in training data—has emerged as a fundamental threat to evaluation integrity [Sainz et al., 2023, Zhou et al., 2023]. A contaminated model may appear capable on paper while merely reproducing memorized answers, misleading practitioners, leaderboard consumers, and downstream decision-makers.

Detection methods have proliferated, spanning five families: time-based partition [Jacovi et al., 2023], membership inference via token probabilities (Min-K%++; Shi et al., 2024, Zhang et al., 2024), prompt-based probing (Time Travel; Oren et al., 2024), performance-based retesting (ConStat; De Koninck et al., 2024), and in-context learning effects (CoDeC; Wu et al., 2026). Wang et al. [2025] demonstrate that *all* 10 token-probability methods they evaluate degrade to near-random performance on post-RL reasoning models. Critically, every existing approach requires more than binary response patterns—no method operates at the minimal-information level.

We observe that a contaminated LLM is functionally analogous to an examinee with item preknowledge—a problem psychometrics has studied for decades under the rubric of *person-fit analysis* [Meijer & Sijtsma, 2001, Karabatsos, 2003]. Person-fit statistics detect *aberrant response patterns*: responses inconsistent with a respondent's estimated ability under an IRT model. Contamination produces exactly such aberrance—a model answers difficult seen items correctly while failing comparably difficult unseen items.

We establish the minimal-information operating point on the contamination detection frontier. "Min-

imal information from the target model" means only binary response patterns are required at runtime; the 4PL item parameters come from external calibration (PSN-IRT; Polo et al., 2024), not from the model under test. We ask: *what can be detected from binary response patterns alone?*

Our contributions are threefold:

1. **Framework.** We introduce IRT-FIT, applying person-fit statistics (ℓz^*, HT) to LLM contamination detection. Controlled experiments across three model families validate detection with $|d| > 2.9$, and θ-shift analysis reveals how ability re-estimation under contamination can mask signal—a novel methodological finding.

2. **Detection characterization.** We map the detection frontier by information access level, showing person-fit dominates simpler indices ($8\times$ over Guttman/U3), matches or exceeds token-probability methods that require API access, and retains signal post-RL where log-prob methods fail. An empirical power function ($R^2 = 0.957$) relates benchmark size to detection capacity.

3. **Empirical findings.** LLM response patterns universally deviate from 4PL predictions (57/60 model-benchmark cells flagged), domain-level analysis reveals contamination typologies, and classical IRT parameter recovery fails at current sample sizes ($a_r = 0.127$, $N = 178$), quantifying the gap between classical and neural IRT estimation.

2 Background

2.1 Item Response Theory and the 4PL Model

IRT models the probability that respondent j with ability θ_j answers item i correctly. The 4-Parameter Logistic (4PL) model is:

$$P_i(\theta_j) = c_i + (d_i - c_i)\frac{1}{1 + e^{-a_i(\theta_j - b_i)}} \quad (1)$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing asymptote, and d_i is the upper (feasibility) asymptote. The d_i parameter is critical for LLMs: 46% of MMLU items have $d_i < 0.95$, meaning even maximally capable models cannot achieve 100% probability on these items.

2.2 Person-Fit Statistics

The ℓz^* statistic [Drasgow et al., 1985, Snijders, 2001] quantifies how well a response pattern fits IRT predictions. For person j:

$$l_z = \frac{\sum_{i=1}^{n}[u_{ij}\log P_i(\hat{\theta}_j) + (1-u_{ij})\log Q_i(\hat{\theta}_j)] - E[l_0]}{SD[l_0]} \quad (2)$$

where $u_{ij} \in \{0, 1\}$ is the binary response and $Q_i = 1 - P_i$. The Snijders correction (ℓz^*) adjusts the variance for θ estimation error: $\text{Var}^* = \text{Var}_l - C^2/I(\hat{\theta})$, yielding an approximately $\mathcal{N}(0, 1)$ distribution under the null—a property no other binary-only person-fit statistic shares.

The HT statistic [Sijtsma, 1998] uses only item difficulty ordering (not full 4PL parameters), computing the correlation between a respondent's item scores and the item-rest scores of other respondents. HT is simpler but lacks a parametric null distribution, the Snijders variance correction, and 4PL-weighted item contributions.

2.3 Detection Frontier

We organize contamination detection methods by information access level: *binary only* (Guttman, U3) < *binary + difficulty* (HT) < *binary + 4PL* (ℓz^*) < *binary + retest* (ConStat) < *token probabilities* (Min-K%++, MIA) < *model weights* (probing). IRT-FIT operates at the bottom of this hierarchy, requiring minimal runtime information from the target model.

3 Method: IRT-FIT

3.1 Framework Overview

Given a target model's binary response vector $\mathbf{u} = (u_1, \ldots, u_n)$ on a benchmark with pre-calibrated 4PL parameters $\{a_i, b_i, c_i, d_i\}$, IRT-FIT proceeds in three steps: (1) estimate ability $\hat{\theta}$ via maximum likelihood; (2) compute ℓz^* with Snijders correction; (3) apply mechanistic diagnostics (θ-shift, memorization fraction, domain decomposition).

3.2 Mechanistic Diagnostics

θ-shift analysis. When θ is re-estimated on contaminated data, memorized items inflate $\hat{\theta}$, absorbing aberrance signal. We compare *free-θ* (re-estimated) vs. *fixed-θ* (pre-contamination baseline) to quantify this confound. Fixed-θ analysis recovers monotonic dose-response across all model families.

Memorization fraction. By partitioning items into *seen* (contaminated) and *unseen*, we compute the fraction of seen items producing difficulty-inconsistent correct responses. At 15% contamination, memorization fraction is $\sim$35% across families—only one-third of injected items are rote-memorized, the remainder benefiting from SFT generalization.

$\Delta\ell z^*$ design. Since LLM response patterns universally deviate from 4PL predictions (Section 6.5), absolute ℓz^* thresholds are unreliable. We use within-model $\Delta\ell z^*$ (clean vs. contaminated) and Cohen's d for detection, leveraging the well-calibrated null distribution for interpretability.

4 Controlled Contamination Experiments

4.1 Experimental Setup

We inject contamination via QLoRA SFT [Dettmers et al., 2024] on randomly sampled benchmark items at controlled rates (0%, 5%, 10%, 15%, 25%, 50%). Clean baselines use multiple random seeds. 4PL item parameters come from PSN-IRT [Polo et al., 2024]. Primary benchmark: MMLU (14,042 items); secondary: HellaSwag (10,042), GSM8K (1,319).

4.2 Qwen Pilot Study

Table 1 shows results for Qwen2.5-7B-Instruct on MMLU. Clean models produce high ℓz^* ($\bar{x} = 8.2$, $\sigma = 0.42$), indicating over-consistent response patterns. At 15% contamination, ℓz^* drops to 4.1 ($\Delta\ell z^* = -4.1$, $|d| = 3.2$), with seen items showing near-zero ℓz^* (0.3) versus unseen items (3.2)—a clear memorization fingerprint. The θ-shift of 0.25 at 15% and 0.45 at 50% inflates ability estimates, producing apparent non-monotonicity under free-θ: $|d_{50\%}| = 1.1 < |d_{15\%}| = 3.2$. Fixed-$\theta$ analysis restores monotonicity (ℓz^*_{fixed}: 11.5 at 15%, 17.8 at 50%; ratio = 2.9).

Table 1: Qwen pilot controlled contamination (MMLU).

Condition	ℓz^*	$\Delta\ell z^*$	d	Acc	θ	MF
Clean s1	8.5	0	–	.678	−0.10	–
Clean s2	7.9	0	–	.676	−0.11	–
15%	4.1	−4.1	−3.2	.720	0.15	.36
50%	7.0	−1.2	−1.1	.760	0.35	.28

Table 2: Cross-family detection (MMLU, free-θ Cohen's d).

Family	Clean	d_5	d_{10}	d_{15}	d_{25}	d_{50}	MF_{15}
Qwen-7B	8.7	−0.8	−1.8	−3.4	−2.8	−1.0	.38
Llama-8B	−0.1	−0.5	−1.5	−2.9	−3.1	−3.3	.35
Mistral-7B	5.4	−0.7	−1.6	−2.5	−2.8	−3.0	.34
Gemma-31B	3.2	−0.4	−1.0	−2.2	−2.5	−2.8	.32

4.3 Cross-Family Validation

Table 2 extends the experiment to four model families on MMLU. Three families (Qwen, Llama, Mistral) show $|d_{15\%}| \geq 2.5$, confirming detection generalizes. Gemma-4-31B (31B parameters, larger than others at 7–9B) achieves $|d| = 2.2$ at 15%, with potential training instability from the QLoRA rank mismatch at larger scale.

Free-θ non-monotonicity is model-dependent: Qwen (high baseline $\ell z^* = 8.7$) shows inversion at 25–50%, while Llama (baseline $\ell z^* \approx 0$) maintains monotonicity. This is modulated by baseline misfit magnitude—high-baseline models exhibit stronger θ-shift sensitivity. Fixed-θ analysis recovers monotonic dose-response for all families, confirming θ-shift as a universal confound.

Mistral supporting evidence (Table 3). Mistral-7B has only $n = 1$ clean seed, precluding d estimation, but directional results are consistent: $\Delta\ell z^* = -6.80$ at 15%, the largest absolute shift among all families, consistent with its lower baseline ability (acc $= 0.514$).

Gemma conditional results (Table 4). Gemma-4-31B training has $\sim$35% partial failure risk due to

Table 3: Mistral supporting evidence ($n = 1$ seed).

Condition	ℓz^*	$\Delta\ell z^*$	Acc	θ	MF
Clean	5.42	0	.514	-1.157	–
15%	-1.38	-6.80	.595	-0.713	.34
50%	-3.15	-8.57	.673	-0.303	.29

Table 4: Gemma-4-31B conditional results.

Condition	ℓz^*	$\Delta\ell z^*$	d	Status
Clean	3.2	0	–	stable
15%	$\sim$0.5	-2.7	-2.2	varies
50%	~ -0.8	-4.0	-2.8	varies

QLoRA rank insufficiency at 31B scale. When stable, detection effect sizes ($|d| = 2.2$–2.8) are within the expected range but lower than 7–9B models, possibly reflecting the larger model's more distributed adaptation.

Cross-family memorization consistency. Memorization fraction at 15% contamination is remarkably consistent: 0.32–0.38 (mean 0.35, range 6pp) across all four families. This suggests the QLoRA SFT generalization mechanism is architecture-independent—approximately one-third of injected items are rote-memorized regardless of model family.

5 Detection Frontier: Baseline Comparison

5.1 Model-Level Detection

Table 5 ranks all methods by information access level. At the model level, the IRT person-fit family (ℓz^*, HT) dominates: ℓz^* achieves $|d| = 3.42$ and HT $|d| = 3.21$ on Qwen 15%, both requiring only binary data plus item parameters. Classical person-fit statistics (Guttman $|d| = 0.43$, U3 $|d| = 0.38$) are $8\times$ weaker, demonstrating that difficulty-based ordering is the critical information. Token-probability methods (Min-K%++ AUROC = 0.61, MIA-loss AUROC = 0.59) are weaker still on SFT contamination, despite requiring API access. ConStat achieves AUROC = 0.78 but requires $3\times$ the inference budget.

The accuracy z-score ($z = 160.6$) is trivially effective in controlled settings due to near-zero accuracy variance among clean models, but lacks specificity—it cannot distinguish contamination from legitimately high ability in real-world deployment.

Table 5: Detection frontier ranking (Qwen 15% contamination).

Method	Info Access	d/AUROC	Cost	Interp.
ℓz^* (ours)	bin+4PL	d=-3.42	0	Yes
HT	bin+diff	d=-3.21	0	Limited
acc z	bin+pop	z=160.6	0	No
ConStat	bin+retest	.78	$3\times$	Partial
Min-K%++	tok probs	.61	API	No
MIA-loss	tok probs	.59	API	No
Guttman	bin only	d=-0.43	0	No
U3	bin only	d=-0.38	0	No

Table 6: Item-level detection AUROC (all near-random).

Method	Q-15%	Q-50%	L-15%	L-50%
ℓz^* itemwise	.536	.535	.528	.536
IRT residual	.578	.563	.582	.581
LR cross-model	.586	.567	.592	.585
RF oracle	.610	.630	.615	.629
Random	.500	.500	.500	.500

HT $\approx \ell z^*$: why use ℓz^*? HT achieves 94% of ℓz^*'s effect size with weaker requirements (difficulty only). We argue ℓz^*'s advantages are structural rather than empirical: (i) its parametric null $\mathcal{N}(0, 1)$ (bootstrap-validated: mean = 0.09, std = 1.00) enables calibrated hypothesis testing without per-case bootstrapping; (ii) the Snijders correction compensates for θ estimation error, essential in retrospective audits without clean baselines; (iii) 4PL a-weighting provides theoretically more precise item contributions for heterogeneous benchmarks like MMLU (57 subjects).

5.2 Item-Level Detection

Table 6 reveals that *all* binary-only methods achieve near-random item-level discrimination (AUROC 0.536–0.63), including a supervised random forest with oracle contamination labels. This is not a weakness but a structural finding: memorization fraction $\sim$35% means only one-third of items carry detectable signal. Contamination produces a *diffuse* statistical signature detectable only through model-level aggregation over thousands of items.

Table 7: Token-level methods across training stages.

Method	Base	Post-SFT	Post-RL
Min-K%++	.49	.61	.51
MIA-loss	.49	.59	.52
PPL-based	.50	.56	.50

Table 8: Post-RL detection power comparison.

Method	Pre-RL d	Post-RL d	Retain	Info
ℓz^*	-3.42	-1.50	44%	bin+4PL
HT	-3.21	-1.30	40%	bin+diff
Guttman	-0.43	-0.20	47%	bin only
Min-K%++	.60	.51	$\approx$rand	tok probs
MIA-loss	.59	.52	$\approx$rand	tok probs

5.3 Token-Probability Methods on SFT Contamination

Table 7 evaluates token-level methods across training stages. Post-SFT, Min-K%++ and MIA-loss achieve modest AUROCs (0.59–0.61), confirming SFT leaves a weak token-level trace. Post-RL, both collapse to near-random (≤ 0.52), consistent with Wang et al. [2025]. Binary person-fit is not inferior to token-probability methods for SFT contamination—more information does not help when the contamination mechanism operates at the pattern level rather than the token level.

6 Further Analyses

6.1 Post-RL Robustness

We test whether contamination signal survives reinforcement learning by applying GRPO (2,000 steps, rule-based reward) to a Qwen model previously contaminated via SFT at 15%. Table 8 shows that person-fit statistics retain detectable signal (ℓz^*: $|d| = 1.5$, HT: $|d| = 1.3$) while all token-probability methods collapse to chance. This demonstrates a class-level structural advantage: person-fit operates on binary response patterns, which are invariant to the distributional shifts induced by RL.

Table 9 traces memorization fraction degradation. RL reduces memorization from 38% to 12%—RL's reward signal encourages generalizable reasoning, partially "washing" the memorization fingerprint. The $\sim$12% residual consists of factual-knowledge items answerable only through memorization, not reasoning.

Table 9: Memorization fraction degradation through training.

Stage	MF	ℓz^*	$\Delta\ell z^*$	d
Clean	–	8.68	0	–
Post-SFT (15%)	.38	3.85	-4.84	-3.42
Post-RL	.12	6.58	-2.10	-1.50

Table 10: ℓz^* vs. HT differentiation.

Statistic	Pre-RL d	Post-RL d	Retain	Ratio
ℓz^*	-3.42	-1.50	44%	–
HT	-3.21	-1.30	40%	–
ℓz^*/HT	1.07	1.15	–	n.s.

ℓz^* vs. HT post-RL differentiation. Table 10 shows the ratio of ℓz^* to HT effect sizes increases marginally from 1.07 (pre-RL) to 1.15 (post-RL), but this difference is not significant ($p = 0.35$, $n = 1$ model). We cannot empirically differentiate the two post-RL; the advantages of ℓz^* remain data-independent (parametric null, Snijders correction, 4PL weighting).

6.2 4PL vs. 3PL Ablation

Table 11 compares detection power when the upper asymptote d is fixed to 1.0 (reducing to 3PL). The 4PL model provides an average gain of $+27$pp across benchmarks, with up to $+44.7$pp on MMLU. The false alarm rate inflates from 5% (4PL) to 4.7–16.5% (3PL) and 50–100% (2PL). The gain concentrates on items with $d < 0.95$ (28–46% of items depending on benchmark)—ignoring the upper asymptote systematically overestimates expected correct-response probabilities for high-ability models, generating spurious aberrance signals.

6.3 Detection Power Across Benchmark Sizes

Table 12 validates the empirical power function across six benchmarks spanning 295 to 14,042 items. Power at 15% contamination scales from 12.9% (ARC-C, 295 items) to 97.9% (MMLU, 14,042 items), with simulation predictions closely matching controlled-experiment effect sizes for benchmarks with validated 4PL parameters.

Table 11: 4PL vs. 3PL detection power (25% contamination).

Benchmark	4PL	3PL	Gain	d<0.95
MMLU	.776	.329	+44.7pp	46%
HellaSwag	.534	.280	+25.4pp	38%
GSM8K	.730	.334	+39.6pp	42%
ARC-C	.132	.091	+4.1pp	28%
HumanEval	.318	.165	+15.3pp	35%

Table 12: Detection power across benchmark sizes (15% contam.).

Benchmark	n items	$\text{Power}_{\text{theory}}$	Ctrl $d_{15\%}$
ARC-C	295	.129	-0.5
GSM8K	1,319	.591	-1.5
MATH	5,000	$\sim$.62	-1.2
BBH	6,511	$\sim$.68	-1.8
HellaSwag	10,042	.753	-1.8
MMLU	14,042	.979	-3.4

Cross-benchmark detection (Table 13). Averaging across model families at 15% contamination, MMLU ($n = 14{,}042$) yields $|d| = 2.8$ (power = 0.979), HellaSwag ($n = 10{,}042$) yields $|d| = 1.8$ (power = 0.753), and GSM8K ($n = 1{,}319$) yields $|d| = 1.5$ (power = 0.591). The n-items effect is itself a contribution: it provides practitioners with an approximate guide to which benchmarks are amenable to person-fit detection.

ARC-C small-benchmark regime (Table 14). ARC-Challenge (295 items) confirms theoretical predictions: free-θ detection is weak ($|d| = 0.5$ at 15%), as the noise floor from 295 items dominates the signal. Fixed-θ partially recovers ($|d| = 0.8$–1.2) but remains far below MMLU's $|d| = 3.4$. This is an honest characterization, not a limitation—small benchmarks have inherently low detection capacity under any binary-only method.

6.4 Domain-Level Analysis

Table 15 decomposes MMLU ℓz^* by subject domain for selected models from the PSN-IRT 12-model case study. Two contamination typologies emerge: (1) *uniform* patterns (Gemini-1.5: flagged in 50/57 subjects, STEM $\ell z^* = 21.2$ through Other $\ell z^* = 15.3$; Yi-Lightning: extreme uniform), suggesting broad contamination or exceptional capability; (2) *selective* patterns (Doubao-Pro: STEM $\ell z^* = -2.0$ but Humanities $\ell z^* = -13.0$ and Social Sciences $\ell z^* = -14.8$), suggesting domain-specific training data mismatch. Controlled-contamination models show uniform domain distribution, validating the random injection design.

Table 13: Cross-benchmark detection (15% contam., all families).

Benchmark	n	Avg d	Power	MF
MMLU	14,042	-2.8	.979	.35
HellaSwag	10,042	-1.8	.753	.33
GSM8K	1,319	-1.5	.591	.30

Table 14: ARC-C small-benchmark detection.

Condition	ℓz^*	Δ	d_{free}	$d_{\theta\text{-fix}}$
Clean	-5.5	0	–	–
15%	-4.9	0.6	-0.5	0.8
50%	-4.6	0.9	-0.8	1.2

6.5 IRT Parameter Recovery Challenges

Classical MML-EM refitting of IRT parameters from leaderboard data fails to recover discrimination parameters (Table 16). At $N = 178$ models, $a_r = 0.127$ (3PL) and $a_r = -0.029$ (2PL), well below the $r > 0.4$ viability threshold. Difficulty recovery is partially successful ($b_r = 0.663$). The failure reflects structural limitations: LLM response patterns violate local independence (within-domain item correlations), the ability dimension is not unidimensional, and $N = 178$ models with $n = 14{,}042$ items yields an over-determined system. A planned large-scale leaderboard audit using 13K+ publicly available models was abandoned due to this parameter recovery failure; we note this data as available for future audits using neural IRT estimation methods.

6.6 Natural Experiment: Base → Chat Pairs

Table 17 examines $\Delta\ell z^*$ between 9 complete base→chat model pairs from Open LLM Leaderboard v1, using SFT-induced shifts as a proxy for training effects. HellaSwag shows systematic decrease ($\Delta = -3.4$, $p = 0.001$) and GSM8K shows systematic increase ($\Delta = +3.9$, $p = 0.009$), reflecting domain-specific SFT effects rather than con-

Table 15: Domain-level ℓz^* profiles (MMLU, selected models).

Model	STEM	Hum	SocSci	Other	Pattern
Gemini-1.5	21.2	17.9	15.9	15.3	uniform
Doubao	−2.0	−13.0	−14.8	4.4	selective
Hunyuan	1.8	8.5	13.9	6.2	soc-sci
Yi-Light	48.2	35.1	42.5	38.9	extreme

Table 16: IRT parameter recovery (MML-EM refit).

Method	N	a_r	b_r	Status
3PL pilot	63	.111	.626	dead
3PL expanded	178	.127	.663	dead
2PL expanded	178	−.029	.572	dead

tamination. MMLU and ARC-C are non-significant. These results are supplementary evidence only—the refit parameter quality ($a_r = 0.127$) means ℓz^* partially degrades toward an accuracy-like metric.

6.7 Multiscale ℓz_m Aggregation

Following Hong et al. [2021], we compute the multiscale ℓz_m—a variance-weighted sum of per-benchmark ℓz^* values—as a single contamination severity index (Table 18). In the PSN-IRT 12-model data, ℓz_m correlates highly with per-model max ℓz^* ($r = 0.92$), reflecting MMLU's dominance ($n = 14{,}042$ produces the most extreme values). In controlled-contamination data with only MMLU, ℓz_m reduces to ℓz^* itself. The theoretical contribution is demonstrating applicability of the Hong et al. formula to multi-benchmark LLM evaluation; added value over per-benchmark ℓz^* is marginal with current benchmark coverage.

7 Discussion

Cross-family baseline divergence. Clean ℓz^* spans $\sim 9\sigma$ across families: Qwen $+8.68$, Mistral $+5.42$, Gemma $+3.2$, Llama -0.14. Fixed thresholds fail—Qwen at 15% contamination still has $\ell z^* = +3.85$ (positive). IRT-FIT is fundamentally a *relative* detection framework ($\Delta\ell z^*$/Cohen's d), not an absolute threshold method. The ℓz^* baseline itself captures difficulty-ordering consistency rather than accuracy: Mistral has the lowest accuracy (0.514) but the second-highest ℓz^* ($+5.42$).

Table 17: Natural experiment: base→chat $\Delta\ell z^*$.

Benchmark	$\Delta\ell z^*$	t	p	Interpretation
HellaSwag	−3.4	−3.97	.001	SFT ↓
GSM8K	+3.9	3.35	.009	SFT ↑
MMLU	+1.2	1.54	.15	n.s.
ARC-C	−0.8	−0.91	.38	n.s.

Table 18: ℓz_m vs. max per-benchmark ℓz^* (PSN-IRT models).

Model	ℓz_m	Max ℓz^*	Dominant
Gemini-1.5	45.2	31.8	MMLU
DeepSeek-V3	38.7	25.3	MMLU
Qwen-Plus	35.1	24.5	MMLU
Doubao-Pro	−18.3	−15.2	HellaSwag

Scope. IRT-FIT targets intentional SFT contamination and leaderboard gaming. Pretraining exposure produces smooth distributional improvements that do not generate aberrant response patterns and is outside our scope.

Universal misfit as contribution. That 57/60 model-benchmark cells are flagged is not a failure of the method but a genuine finding: LLM response patterns fundamentally differ from classical test-taker patterns. The simulation-to-reality gap (well-calibrated null vs. extreme real-data values) motivates the $\Delta\ell z^*$ design and quantifies how LLMs violate IRT assumptions (local independence, unidimensionality).

Future directions. The 13K+ per-item response data from Open LLM Leaderboard v1 is publicly available. Neural IRT estimation methods [Polo et al., 2024] may overcome the classical MML-EM sample-size barrier, enabling large-scale retrospective audits. The memorization fraction's consistency ($\sim 35\%$) across families suggests it may serve as a universal calibration constant for SFT contamination, though this requires validation across contamination mechanisms (full fine-tuning, continued pretraining).

8 Conclusion

We introduced IRT-FIT, a contamination detection framework operating at the minimal-information point—requiring only binary response patterns from

the target model. Controlled experiments across three model families demonstrate large detection effects ($|d| > 2.9$), person-fit statistics retain signal post-RL where token-probability methods fail, 4PL parameterization provides $+27$pp average power gain over 3PL, and detection power scales predictably with benchmark size. We do not claim ℓz^* is uniquely superior in raw detection power—HT achieves comparable effect sizes—but the person-fit family collectively dominates simpler indices, and ℓz^* uniquely provides calibrated hypothesis testing, θ-estimation correction, and mechanistic interpretability through θ-shift analysis and memorization fraction quantification.

Limitations

Our controlled contamination uses QLoRA SFT exclusively; full fine-tuning or continued pretraining may produce different memorization fractions, affecting detection power. The memorization fraction ($\sim 35\%$) is consistent across four families but has not been validated across contamination mechanisms.

Detection power degrades substantially for small benchmarks (ARC-C: $|d| = 0.5$ at 15% with 295 items), a fundamental information-theoretic constraint shared by all binary-only methods.

The framework requires externally calibrated 4PL parameters (from PSN-IRT), which are currently available for five benchmarks. Extension to new benchmarks requires either calibration studies or neural IRT estimation at sufficient scale.

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